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HUJI Applicative Research Report — Week 44, 2026

3 high-potential paper(s) · Weekly · Week 44, 2026 · Generated October 26, 2026

Hebrew University researchers are advancing therapeutic strategies and drug discovery tools with significant commercial potential. One breakthrough involves a neuron-specific gene therapy platform poised to address a range of neurodevelopmental disorders. Another innovation presents a pump-free microfluidic device, offering a simplified and cost-effective solution for pharmaceutical drug screening. Furthermore, a novel metabolic target has been identified for developing new antibacterial treatments, highlighting diverse approaches to critical health challenges.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Rami I Aqeilan** — Drug Discovery, Neuroscience, Genomics, Synthetic Biology
- **Ofra Benny** — Drug Discovery, Medical Device, Diagnostics
- **Emanuel Hanski** — Drug Discovery, Immunology, Proteomics, Clinical

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Neuron-specific WWOX gene therapy produces dose-dependent, durable rescue in a model of WWOX-related epileptic encephalopathy. 31/50

"Neuron-specific gene therapy platform offers durable rescue for severe neurological disorders."

Main researcher: Rami I Aqeilan

The Concern Foundation Laboratories, The Lautenberg Center for Immunology and Cancer Research, Department of Immunology and Cancer Research-IMRIC, Hebrew University-Hadassah Medical School, Jerusalem, Israel.

Published in Molecular therapy. Advances · 2026-09

ABOUT THIS RESEARCH

Researchers developed a targeted gene therapy that replaces the WWOX gene specifically in neurons to treat severe epilepsy and developmental disorders. The study demonstrates that this approach provides long-lasting and dose-dependent improvements in animal models of the disease.

COMMERCIAL ANGLE

The development of a neuron-specific delivery framework provides a platform technology for treating a wide range of neurodevelopmental genetic disorders currently lacking therapeutic options.

COMMERCIALISATION METRICS

Novelty 7/10

The research demonstrates significant translational innovation by moving beyond simple gene replacement to a 'rationally optimized' framework with neuron-specific targeting and dose-dependency. While gene replacement for WWOX is a logical therapeutic approach, the optimized delivery mechanism and the establishment of critical clinical parameters provide high-value, platform-enabling insights.

Commercial Potential 8/10

The study addresses a high-unmet-need orphan disease with a clearly defined therapeutic modality (gene replacement) and provides critical proof-of-concept regarding dose-dependency and durability. The use of a 'rationally optimized framework' suggests a potentially licensable platform technology for WWOX-related pathologies.

Market Size 4/10

The target market is limited to ultra-orphan populations (WOREE and SCAR12 syndromes), which represent extremely small patient cohorts. While the technology demonstrates high clinical utility, the addressable demand is constrained by the rarity of these specific genetic indications.

Tech Readiness 4/10

The research is at the preclinical stage, demonstrating efficacy and dose-dependency in a disease model rather than in humans. While it optimizes critical parameters for translation, it has not yet entered formal IND-enabling studies or clinical trials.

IP Strength 8/10

The development of a 'rationally optimized' neuron-specific framework suggests proprietary engineering of promoters or delivery vectors, which provides strong platform-level patentability beyond the transgene itself. This specificity enhances defensibility by creating a unique therapeutic window and reducing off-target risks compared to generic gene replacement.

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2. A pump-free microfluidic device for integrated multi-functional testing of tumor spheroids. 30/50

"Novel pump-free microfluidic device revolutionizes drug screening with low-cost, plug-and-play organ-on-a-chip."

Main researcher: Ofra Benny

The Institute for Drug Research, The School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in APL bioengineering · 2026-06

ABOUT THIS RESEARCH

Researchers have developed a microfluidic platform that uses passive mechanisms instead of external pumps to study 3D tumor spheroids. This device allows for more integrated and simplified testing of how cancer models react to different environments or drugs.

COMMERCIAL ANGLE

The technology can be commercialized as a low-cost, plug-and-play organ-on-a-chip platform for pharmaceutical companies to improve the accuracy of drug toxicity and efficacy screening.

COMMERCIALISATION METRICS

Novelty 6/10

While the elimination of external pumps simplifies microfluidic architecture, the integration of tumor spheroid testing via passive fluidics is an incremental engineering improvement rather than a paradigm-shifting scientific discovery. The novelty lies in the device's specific configuration for ease of use, but it builds upon well-established principles of capillary-driven or gravity-driven microfluidics.

Commercial Potential 7/10

The device addresses a significant bottleneck in drug discovery (high-throughput 3D modeling) and the 'pump-free' aspect suggests a scalable, cost-effective hardware platform. However, commercial success depends on navigating the highly competitive organ-on-a-chip market and demonstrating clear workflow integration for pharmaceutical R&D.

Market Size 7/10

The technology targets the high-growth organ-on-a-chip and drug discovery market, offering significant value by reducing infrastructure costs and improving high-throughput screening scalability. However, it is a specialized tool within a niche segment of the broader preclinical testing market rather than a broad-scale clinical diagnostic.

Tech Readiness 4/10

The paper describes the fundamental development and proof-of-concept of a microfluidic prototype, which is currently at the laboratory validation stage. While it addresses a significant need in drug testing, it lacks the standardized manufacturing and rigorous biological validation required for near-market or industrial adoption.

IP Strength 6/10

The innovation lies in the physical architecture of a passive microfluidic system, which is patentable, but the lack of a proprietary reagent or biological readout limits its platform defensibility against broader microfluidic competitors.

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3. Group A Streptococcus host-pathogen dual crosstalk. 28/50

"New metabolic target discovered for starving bacterial pathogens, combating antibiotic resistance."

Main researcher: Emanuel Hanski · emanuelh@ekmd.huji.ac.il.
Department of Microbiology and Molecular Genetics, The Institute for Medical Research, Israel-Canada (IMRIC), Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem 9112102, Israel. Electronic address: emanuelh@ekmd.huji.ac.il.
Published in Trends in microbiology · 2026-06

ABOUT THIS RESEARCH

Researchers have discovered that Group A Streptococcus bacteria hijack the human cell's stress response to force the production of the amino acid asparagine, which the bacteria then consume to fuel infection. This mechanism creates a metabolic bridge between host cell stress and bacterial virulence, mimicking similar pathways seen in cancer cells.

COMMERCIAL ANGLE

The discovery of the PERK-eIF2α-ATF4-Asn axis provides a novel metabolic target for developing small-molecule inhibitors to starve bacterial pathogens or disrupt their ability to exploit host resources.

COMMERCIALISATION METRICS

Novelty 8/10
The research identifies a highly specific, previously unknown metabolic hijacking mechanism where an extracellular bacterium exploits a host stress pathway (PERK-eIF2α-ATF4) to drive nutrient biosynthesis. This discovery of a direct 'crosstalk' circuit provides a novel, targetable metabolic vulnerability that parallels oncogenic pathways, offering significant potential for new antimicrobial strategies.

Commercial Potential 6/10
The identification of a specific metabolic axis (ATF4-Asn) provides a clear target for novel antimicrobial drug development, though the transition from mechanistic discovery to a viable therapeutic lead remains speculative.

Market Size 7/10
The research targets Group A Streptococcus, a major global pathogen with high clinical burden, but its most significant market value lies in the potential to translate the ATF4-Asn metabolic axis into oncology therapeutics. While the infectious disease application is substantial, the broader addressable market is expanded by the cross-sector relevance to cancer metabolism and therapy resistance.

Tech Readiness 2/10
The paper describes fundamental mechanistic biology and discovery of a signaling pathway in a laboratory setting. It lacks any mention of therapeutic intervention, drug development, or preclinical validation required to move beyond basic research.

IP Strength 5/10
The paper describes a fundamental biological mechanism (the ATF4-Asn metabolic axis) rather than a specific therapeutic molecule or engineered platform. While the discovery provides a targetable pathway, the core finding is a natural phenomenon which is difficult to patent without specific, non-obvious chemical or biological interventions.

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